Lecture: One-Sample t-Test

From z-Tests to Student's t

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1 Setup and Motivation

Suppose we have an i.i.d. sample:

$$Y_1, Y_2, \ldots, Y_n \sim \mathcal{N}(\mu, \sigma^2).$$

We want to estimate (and test hypotheses about) μ and σ^2.

1.1 Maximum Likelihood Estimates (MLE)

The MLEs are:

$$\hat{\mu} = \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2 \quad \text{(biased)}.$$

In practice we prefer the **unbiased** sample variance (requires $n \geq 2$):

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2.$$

> **Note (n = 1 edge case):** If $n = 1$, then $\hat{\mu} = \bar{y} = y_1$ and $\hat{\sigma}^2 = (y_1 - \bar{y})^2 = 0$, which is degenerate. Always ensure $n \geq 2$.

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2 The z-Test (Known Variance)

Hypotheses:

$$H_0 : \mu = 0 \quad \text{vs} \quad H_1 : \mu \neq 0.$$

Test statistic: The sample mean satisfies

$$\mathbb{E}[\bar{Y}] = \mu, \qquad \mathrm{Var}(\bar{Y}) = \frac{\sigma^2}{n},$$

so after standardising:

$$Z = \frac{\bar{Y} - \mu_0}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1) \quad \text{when } \sigma = \sigma_0 \text{ is known.}$$

The **p-value** for the two-sided test is $P(|Z| \geq z_{\text{obs}})$.

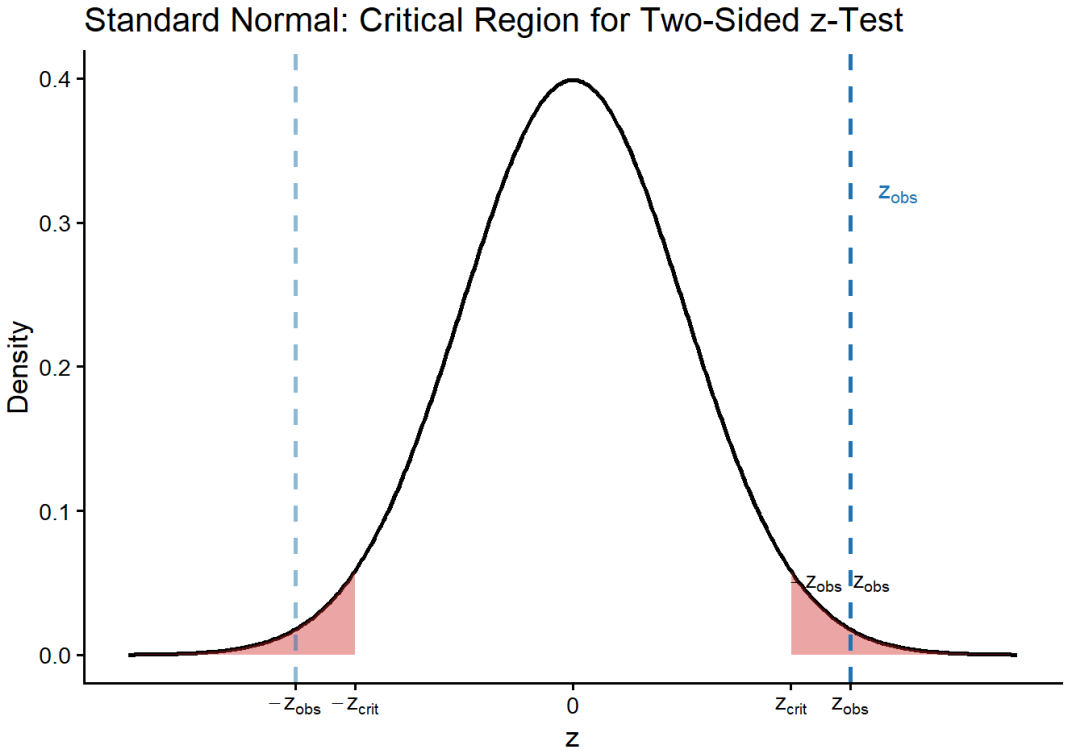

Figure 1: Standard normal distribution with critical region (two-sided, α = 0.05) shaded. The observed test statistic z_obs = 2.5 is shown.

> **Practical limitation:** z-tests are rarely useful because σ^2 is almost never known in practice.

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3 The t-Test (Unknown Variance)

3.1 Plug-in estimator

When σ^2 is unknown, we substitute s^2 for σ^2:

$$T = \frac{\bar{Y} - 0}{\sqrt{s^2/n}} = \frac{\bar{Y}}{s/\sqrt{n}}.$$

This statistic is **no longer** $\mathcal{N}(0,1)$-distributed.

3.2 William Gosset ("Student") and the t-distribution

Gosset showed that if $\bar{Y} \sim \mathcal{N}(\mu, \sigma^2/n)$ and $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2$, then:

$$\boxed{\frac{\bar{Y} - \mu}{s/\sqrt{n}} \sim t_{n-1}}$$

where t_{n-1} denotes the **Student's t-distribution with** $n-1$ **degrees of freedom**.

As $n \to \infty$ (degrees of freedom $\to \infty$), $t_\nu \to \mathcal{N}(0,1)$.

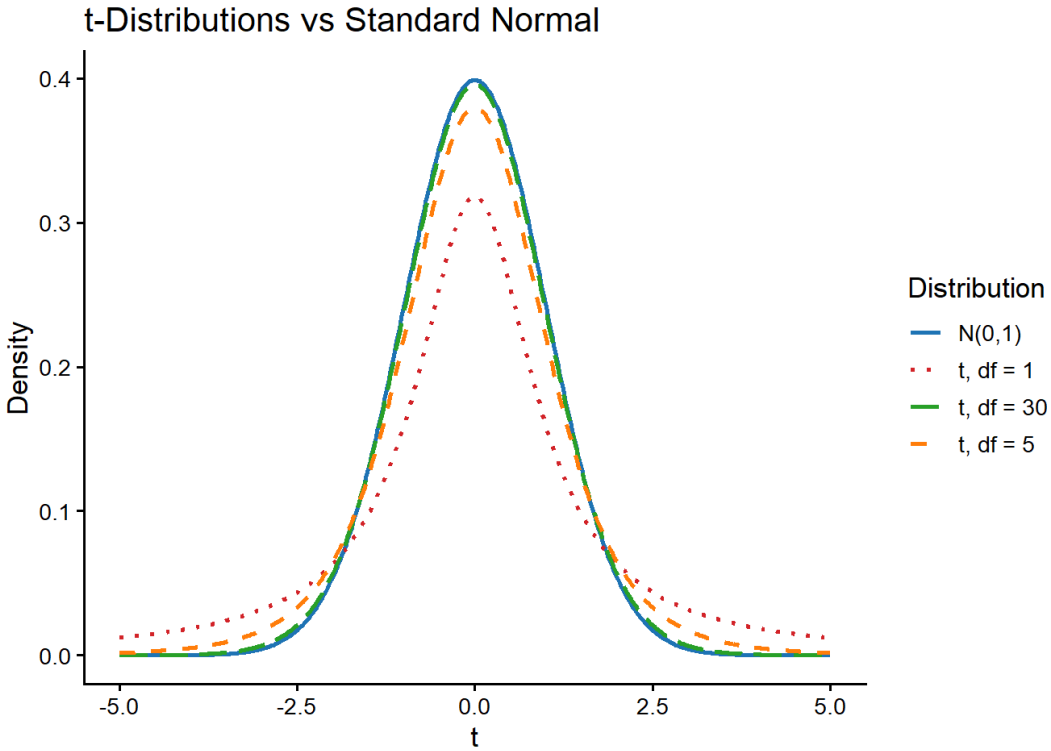

Figure 2: t-distributions for increasing degrees of freedom compared to the standard normal. As df increases, the t-distribution converges to N(0,1), but has heavier tails for small df.

3.3 Computing p-values in R

In R, the t-distribution functions follow the standard `d/p/q/r` naming convention:

Prefix	Function	Purpose
`d`	`dt()`	Density
`p`	`pt()`	CDF / Probability
`q`	`qt()`	Quantile
`r`	`rt()`	Random sample

For a two-sided test the p-value is:

$$p\text{-value} = P(|T| \geq t_{\text{obs}}) = 2 \times P(T \geq |t_{\text{obs}}|) = 2 \times (1 - \texttt{pt}(t_{\text{obs}},\, n-1))$$

3.4 Example

Read the `gpa_study_hours` data set from https://www.openintro.org/data/index.php?data=gpa_study_hours. This dataset has 193 rows and 2 columns. The columns represent the variables `gpa` and `study_hours` for a sample of 193 undergraduate students who took an introductory statistics course in 2012 at a private US university.

We are interested in testing hypothesis about the mean `study_hours`, i.e. if students in this particular class studied more than a specified number of hours, e.g., we can ask if they have studied for more than 15 hours?

```
'data.frame':   193 obs. of  2 variables:
 $ gpa        : num  4 3.8 3.93 3.4 3.2 3.52 3.68 3.4 3.7 3.75 ...
 $ study_hours: num  10 25 45 10 4 10 24 40 10 10 ...

sample size: 193 sample mean: 17.47668 standard error: 0.8212363 t:
3.015799
```

```r
p_upper    <- 1 - pt(t_stat, df = n - 1)   # one-sided (H1: mu > mu_0)
p_twosided <- 2 * p_upper                  # two-sided (H1: mu ≠ mu_0)

cat(sprintf("One-sided p-value  : %.4f\n", p_upper))
```

```
One-sided p-value  : 0.0015
```

```r
cat(sprintf("Two-sided p-value  : %.4f\n", p_twosided))
```

```
Two-sided p-value  : 0.0029
```

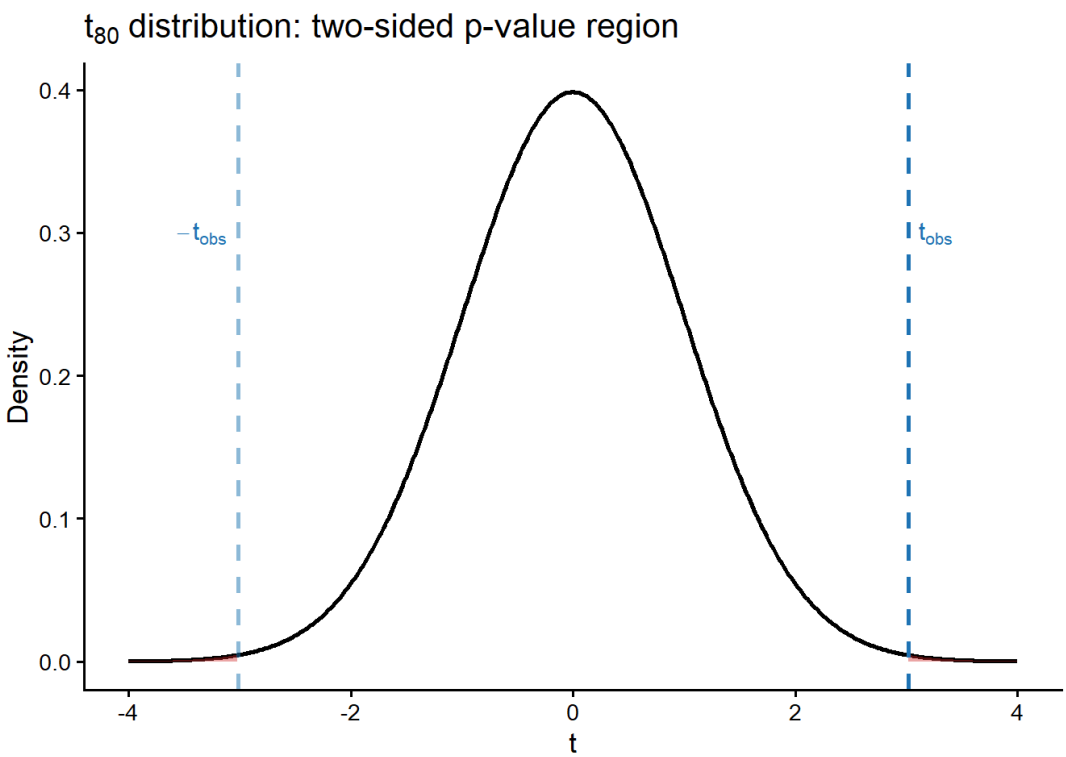

Figure 3: t-distribution with n-1s degrees of freedom. The shaded regions represent the two-sided p-value corresponding to t_stat = 3.015799.

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4 Distribution of s^2: The Chi-Squared Connection

To derive the t-distribution formally, we need the sampling distribution of s^2.

4.1 Chi-squared distribution

If $Y \sim \mathcal{N}(0,1)$, then $Y^2 \sim \chi^2_1$.

More generally, if $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$, then:

$$\sum_{i=1}^{n} Y_i^2 \sim \chi_n^2.$$

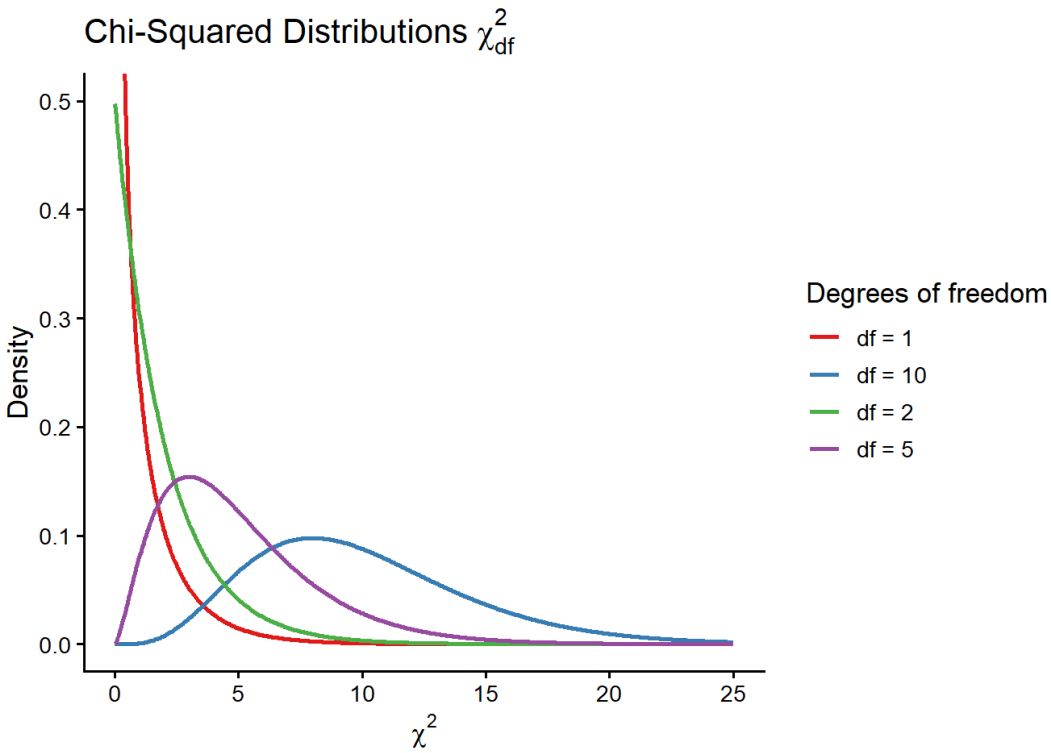

Figure 4: Chi-squared densities for increasing degrees of freedom. The distribution is right-skewed for low df and becomes more symmetric as df grows.

4.2 Decomposition of the total sum of squares

For $Y_i \sim \mathcal{N}(\mu, \sigma^2)$, we know $\sum_{i=1}^{n} \left(\frac{Y_i - \mu}{\sigma}\right)^2 \sim \chi_n^2$.

By adding and subtracting $\bar{y}$:

$$\sum_{i=1}^{n} \left(\frac{Y_i - \mu}{\sigma}\right)^2 = \sum_{i=1}^{n} \frac{(Y_i - \bar{Y})^2}{\sigma^2} + \left(\frac{\bar{Y} - \mu}{\sigma/\sqrt{n}}\right)^2 = \frac{(n-1)s^2}{\sigma^2} + \underbrace{\left(\frac{\bar{Y} - \mu}{\sigma/\sqrt{n}}\right)^2}_{\sim \chi_1^2}.$$

The cross-term $\sum_i (y_i - \bar{y})(\bar{y} - \mu) = 0$ by construction, so by independence of the two components:

$$\boxed{\frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2}$$

Equivalently:

$$s^2 \sim \frac{\sigma^2}{n-1} \cdot \chi_{n-1}^2, \qquad \sigma^2 \sim \frac{(n-1)s^2}{\chi_{n-1}^2}.$$

5 Summary: The One-Sample t-Test

Setup: $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, both μ and σ^2 unknown.

Hypotheses:

$$H_0 : \mu = \mu_0 \quad \text{vs} \quad H_1 : \mu \neq \mu_0 \quad \text{(two-sided)}.$$

Combining results: Since $\bar{Y} \sim \mathcal{N}(\mu, \sigma^2/n)$ and $\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$ are **independent**, their ratio yields:

$$\bar{Y} \sim t_{n-1}\left(\mu, \frac{s^2}{n}\right), \qquad T = \frac{\bar{Y} - \mu_0}{s/\sqrt{n}} \sim t_{n-1} \quad \text{under } H_0.$$

p-values:

$$p\text{-value} = \begin{cases} P(T > t_{\text{obs}}) & \text{if } H_1 : \mu > \mu_0, \\ P(|T| > t_{\text{obs}}) & \text{if } H_1 : \mu \neq \mu_0. \end{cases}$$

95% Confidence interval:

$$\bar{Y} \pm t_{n-1,\,0.975} \cdot \frac{s}{\sqrt{n}} \approx \bar{Y} \pm 1.96 \cdot \frac{s}{\sqrt{n}} \quad \text{(large } n\text{)}.$$

In R:

```r
# Simulate some data and run a one-sample t-test
set.seed(42)
y <- rnorm(30, mean = 0.3, sd = 1)

result <- t.test(y, mu = 0)
print(result)
```

```
    One Sample t-test

data:  y
t = 1.6086, df = 29, p-value = 0.1185
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 -0.1000484  0.8372220
sample estimates:
mean of x
0.3685868
```

5.1 Caveat: Gaussianity assumption

The exact result $\frac{\bar{Y}-\mu}{s/\sqrt{n}} \sim t_{n-1}$ holds **only** when $Y_1, \ldots, Y_n$ are Gaussian. For non-Gaussian data, alternatives include:

- **Nonparametric tests** (e.g., Wilcoxon signed-rank test).

- **Monte Carlo / permutation approaches** to obtain the sampling distribution of the test statistic empirically.

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6 Problem: Gaussianity Assumption and Alternatives

```
B <- 10000
n <- 30
t_stats <- replicate(B, {
  y_sim <- rnorm(n, mean = 0, sd = 1)
  t.test(y_sim, mu = 0)$statistic
})

x_seq <- seq(-4, 4, length.out = 300)
theory <- data.frame(x = x_seq, y = dt(x_seq, df = n - 1))

ggplot(data.frame(t = t_stats), aes(x = t)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 60, fill = "#aec7e8", colour = "white") +
  geom_line(data = theory, aes(x, y), colour = "#d62728", linewidth = 1) +
  labs(x = "t-statistic", y = "Density",
       title = expression("Sampling Distribution of T under H"[0] ~ "(Monte
```

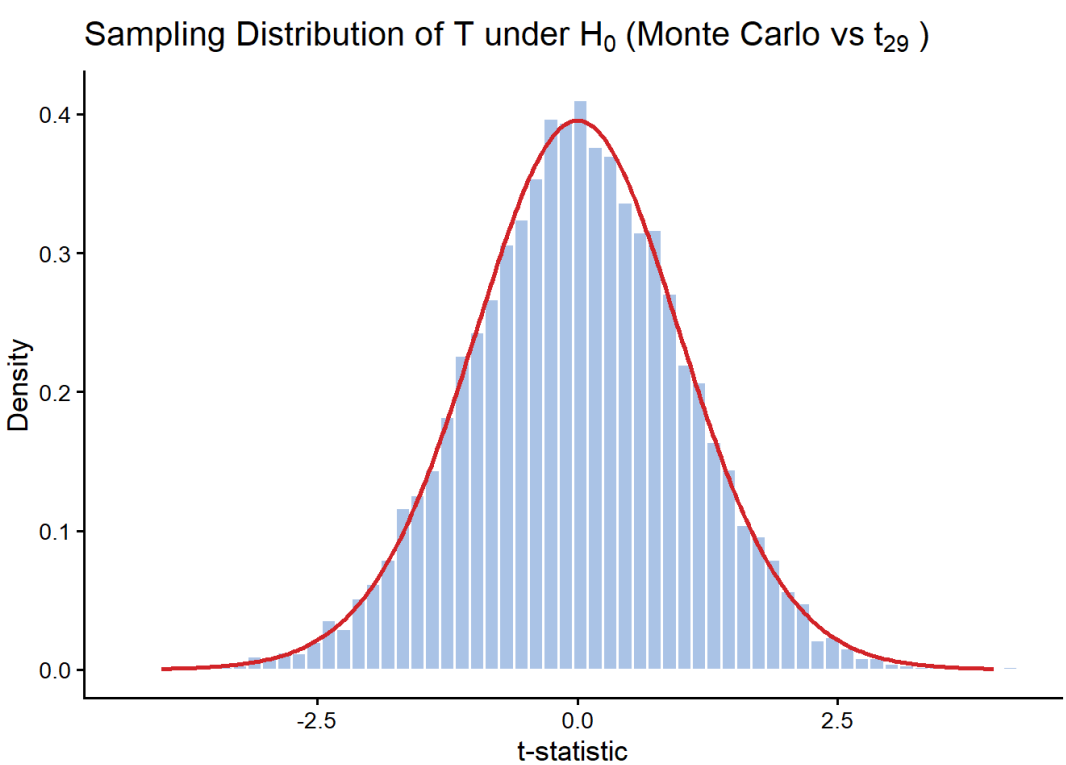

Figure 5: Empirical sampling distribution of the t-statistic under the null, obtained via Monte Carlo simulation (n = 30, B = 10,000 replicates), overlaid with the theoretical t_{29} density. The close agreement illustrates how the t-distribution arises in practice.